Case Study: Operationalizing Growth via Behavioral Data Science at BauData Chile

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Executive Summary

This project integrates two distinct analytical dimensions—an external UX/Behavioral audit and internal data science modeling—under the **Flywheel Framework**. The goal was to diagnose why 67.92% of user sessions ended in friction zones despite a robust data offering, proposing a strategy based on aligning user intent with product value.

1. Strategic Approach: Integration via Flywheel

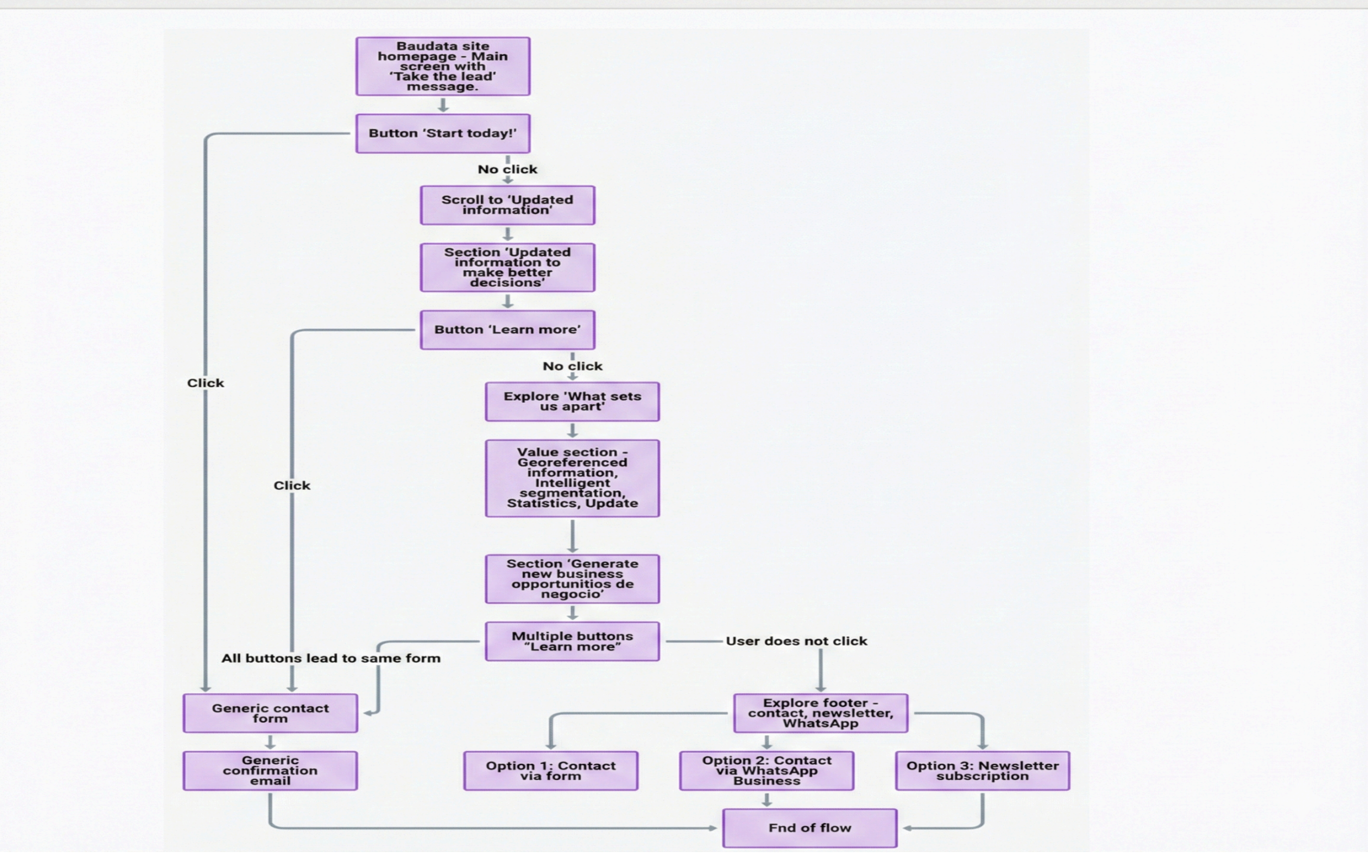
Given BauData's nature as a data-as-a-service platform, the analysis was structured to cover the entire user lifecycle. Although the data sources for the attraction and engagement phases differ, they were unified through **Flywheel logic**:

- **Attraction & Initial Interaction (External Audit):** Qualitative and semantic analysis of the public landing page to understand the value promise and search intent alignment.
- **Engagement & Retention (Internal Analysis):** Quantitative modeling using Machine Learning on real behavioral data captured via telemetry.

2. Behavioral Audit: The Root of Friction (External Analysis)

I conducted an independent analysis of the platform's public touchpoints, identifying a structural flaw in user acquisition:

- **Intent Misalignment:** The landing page acted as a "monolithic funnel," forcing diverse user profiles—from technical roles seeking raw data to commercial roles seeking demos—into the same generic contact form.
- **Semantic Mismatch:** There was a clear disconnect between high-intent SEO keywords (e.g., "Social Housing regulations") and the landing page's visual response, triggering premature bounces before users reached the product's core value.



3. Technical Approach: Validation via Machine Learning (Internal Analysis)

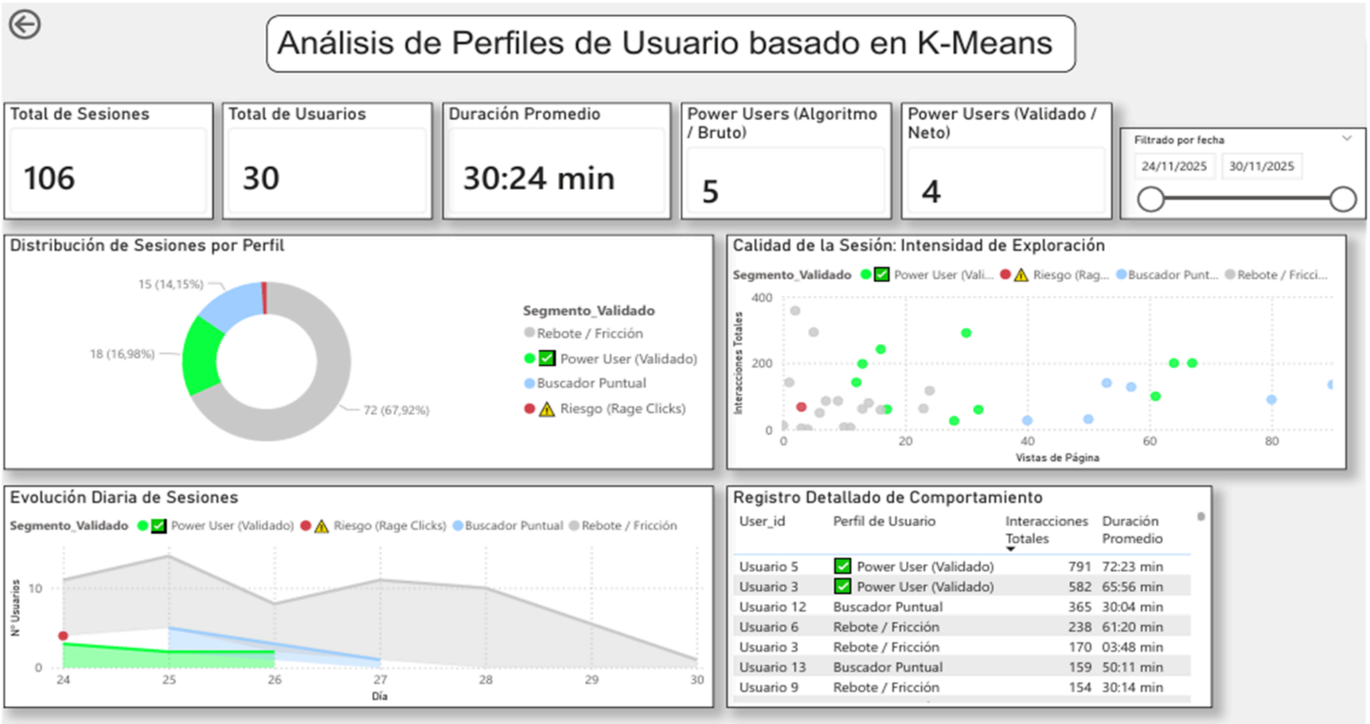
To validate whether the audit findings impacted actual usage, internal platform telemetry was processed:

- **Data Ingestion:** Analyzed 1,623 unique events captured via **PostHog**.
- **Modeling:** Implemented a **K-Means Clustering (k=3)** algorithm to segment users based on their actual behavior once inside the platform.

4. Findings: Anatomy of the Session

The model confirmed that the misalignment detected in the audit translated into three distinct usage profiles:

1. **Power Users (14.15%):** The high-value core. Users who successfully navigate initial barriers to find deep data insights.
2. **Targeted Searchers (16.98%):** High-efficiency transactional users who enter for a specific data point and exit quickly.
3. **Friction Zone (67.92%):** The majority segment. Long-duration sessions with low effective interaction, where users appear "trapped" trying to resolve a need that the interface does not immediately facilitate.



5. Strategic Roadmap: Operationalizing Growth

The connection between both analyses allows for a clear, data-driven roadmap:

- **UX/SEO Alignment:** Redesigning the landing page to ensure the "user journey" begins without semantic friction.
- **Behavioral Funnel Analysis:** Utilizing telemetry to pinpoint the exact drop-off points where the 67% of "Friction" users abandon the flow.
- **Model Iteration (Roadmap):** A second phase of **Feature Engineering** is planned to apply differentiated weights to high-conversion events (such as PDF downloads) to further refine the Power User segment.

This case study demonstrates the ability to bridge qualitative product analysis with the technical rigor of data science.